

```
def goldbach(n):
    for p in prime_range(2, n):
        if is_prime(n - p):
            return (p, n - p)
    return None

print(goldbach(123456780))
```

Another famous unsolved problem in number theory is the Twin Prime Conjecture, first formally stated in the 19th century. It states that there exist infinitely many pairs of prime numbers that differ by two. Such pairs are known as twin primes. Examples include (3,5), (5,7), (11,13), and (41,43). Although twin prime pairs become increasingly rare as numbers grow larger, millions of such pairs have been discovered through extensive computer searches. However, no mathematical proof has yet established that infinitely many twin prime pairs exist. Using SageMath, we can easily search for twin prime pairs within any specified range and identify the largest twin prime pair found in that range. The simple SageMath program *twinprime49.sage*, shown below, searches for all twin prime pairs less than a specified limit and returns the largest pair found. Notice that the function `is_prime()` used in the program is a highly optimized primality-testing function provided by SageMath, enabling the program to efficiently search for twin prime pairs.

```
def largest_twin_prime(limit):
    largest = None

    for p in prime_range(2, limit):
        if is_prime(p + 2):
            largest = (p, p + 2)
    return largest

limit = 1000000
pair = largest_twin_prime(limit)
print(pair)
```

Another famous unsolved problem in mathematics is the Collatz Conjecture, proposed by the German

mathematician Lothar Collatz in 1937. The conjecture begins with any positive integer. If the number is even, divide it by 2; if it is odd, multiply it by 3 and add 1. Repeating this process generates a sequence of integers. The Collatz Conjecture states that every positive integer eventually reaches the value 1, regardless of the starting number. Despite its remarkably simple statement, no proof or counterexample has yet been discovered. For example, starting with 6 produces the sequence  $6 \rightarrow 3 \rightarrow 10 \rightarrow 5 \rightarrow 16 \rightarrow 8 \rightarrow 4 \rightarrow 2 \rightarrow 1$ , while starting with 11 gives  $11 \rightarrow 34 \rightarrow 17 \rightarrow 52 \rightarrow 26 \rightarrow 13 \rightarrow 40 \rightarrow 20 \rightarrow 10 \rightarrow 5 \rightarrow 16 \rightarrow 8 \rightarrow 4 \rightarrow 2 \rightarrow 1$ . SageMath makes it easy to experiment with this intriguing conjecture. The simple SageMath program *collatz50.sage*, shown below, generates the Collatz sequence for a given positive integer by repeatedly applying the rules of the Collatz Conjecture until the sequence reaches 1.

```
def collatz(n):
    n = ZZ(n)
    seq = [n]

    while n != 1:
        if n % 2 == 0:
            n = n // 2
        else:
            n = 3*n + 1
        seq.append(n)

    return seq
```

```
num = 21

print("Collatz sequence for", num)
print(collatz(num))
```

Notice that the program converts the input into a SageMath integer using the function `ZZ()`. Unlike Python's built-in integer type, SageMath's integer type is backed by the highly optimized GNU Multiple Precision Arithmetic Library (GMP), which is designed for fast arithmetic on arbitrarily large integers. Consequently, the program can

efficiently compute Collatz sequences even for extremely large starting values without encountering integer overflow. Although the Collatz algorithm itself is simple, SageMath's optimized integer arithmetic makes it well suited for large-scale computational experiments related to the conjecture.

Now, let us execute these three programs and examine their output. Figure 1 shows the output produced when the programs are executed. Verify that each program behaves as expected. Confirm that the pair returned by *goldbach48.sage* consists of two prime numbers whose sum equals the given even integer, and that the pair returned by *twinprime49.sage* consists of two prime numbers differing by exactly 2. You may use the SageMath function `is_prime()` to verify the primality of the numbers returned by both programs. Finally, verify that every term in the Collatz sequence generated by *collatz50.sage* follows the rules of the Collatz Conjecture and that the sequence eventually terminates at 1. With that, we conclude our discussion of number theory.

```
(sage) # sage goldbach48.sage
(19, 123456761)
(sage) # sage twinprime49.sage
(999959, 999961)
(sage) # sage collatz50.sage
Collatz sequence for 21
[21, 64, 32, 16, 8, 4, 2, 1]
```

Figure 1: Execution of the Goldbach, Twin Prime, and Collatz programs

As this is the final article in this series, let us conclude by taking a brief tour of a few other fields of mathematics and computer science where SageMath excels. Its scope is remarkably broad, encompassing numerous branches of mathematics, engineering, and scientific computing. While it is impossible to cover them all within the confines of this series, it is worthwhile to highlight SageMath's capabilities in four important areas: numerical analysis, statistics, linear algebra, and calculus.